

NETWORK DESIGN

ANALYSIS OF BUSINESS NEEDS

Chesha is a call rental company that gives vehicles for rent. The company will ensure that the vehicle is rented only by the people who have UAE license. The rent will last for maximum of 6 months. The staff has been facing certain issues with the paper-based system that it uses, as the system is never accurate and therefore has decided to move its business online.

The company requires six standalone PCs for the employees and therefore at least **6 PCs** are required to be connected to the switch, in a single LAN. Further, the roaming laptops would also be connected to the network. The operating systems can be Windows, Linux or MAC. **Windows** operating system seems like a better option because of the high functionality and security of the software. It makes work quite easy with all its features – high efficiency is one of the key features of the software. MAC and Linux are a lot costlier in comparison to Windows.

Most of the components would be in a fixed location indicating the wired network usage. Wired connections are faster and safer and can be used for the company's needs. There are several different types of cables that can be used for the connection such as the copper cable, fiber optics cables, etc. However, **copper cables** seem to be a better option due to the price range. Fiber optics are a lot costlier and harder to implement.

Customer needs to be attracted to the company and for this purpose, free Wi-Fi can be provided to them. This is done by connecting the wireless system to the network. **WLAN** is a perfect option for the purpose since it is a lot cheaper to buy and implement. The security for this Wi-Fi needs to be well - maintained. WPA or WPA2 can be used for this purpose. It provides password security to WLAN. **WPA2** seems better due to its strong encryption and passwords.

Software applications would be required for storing the database – MS access or MySQL can be used for this purpose. **MySQL** seems to be a better option as it supports several different platforms and operating systems. It is open source and can be used by anyone for free. It also has better performance power in comparison to MS Access. Mail and messages transfers would be essential, both within and outside the company, Outlook or Spike can be used for this purpose of interaction. According to me, **Spike** is a better option as it allows setting emails according to the priority and allows calling. **Wave** can be used for sharing and printing invoices as that would also be required once the customer place an order.

A website also needs to be designed for the company. This will help them create their presence in the market as well as attract more customers. **DNS** protocol is used for the domain service of the website. All the website details would be stored on a server. **HTML** and **CSS** can be used for designing the webpage. The website must be accessible through the domain name as well as the IP address. **HTTPS** or HTTP websites can be developed. Https indicates that the website is a lot more secure and safe.

Server is quite essential. It must be able to store the database, customer as well as the invoice application. There can also be two different servers - one for email services - the other for website maintenance. However, since Chelsha is a small company and has a low budget, having all the processes in one server seems to be a better option.

The email services on the server would require a protocol – **SMTP** and **POP3** protocols can be enabled. SMTP, also known as Simple mail transfer protocols, allows the mail transfer between different systems whereas POP3 is used for receiving the mail. The application software will be available in the TCP/IP layer, UDP or TCP protocols can be used for transferring the data packets. UDP is faster but is quite unreliable. On the other hand, TCP is slower but a lot more reliable and secure as it is a connection-oriented protocol. **TCP** seems to be a better choice due to its reliability and security. A protocol is also required for transferring important files - **FTP** or File transfer protocol is a reliable protocol used for transferring files. It is very reliable and secure and can therefore be used for this purpose.

Security must also be kept in mind while sharing the interaction, messages and file transfer and the printing of the data. Firewalls and several other mechanisms such as Antivirus, passwords, packet filters etc. can be used for the safety of the network. **Packet filters** would avoid any suspicious packets getting into the network and thus protect the network as well as the system.

DESIGNING AND DEVELOPING ALTERNATIVE SOLUTIONS TO MEET THE DESIGN BRIEF

Three alternate solutions have been developed for the Chelsha Company.

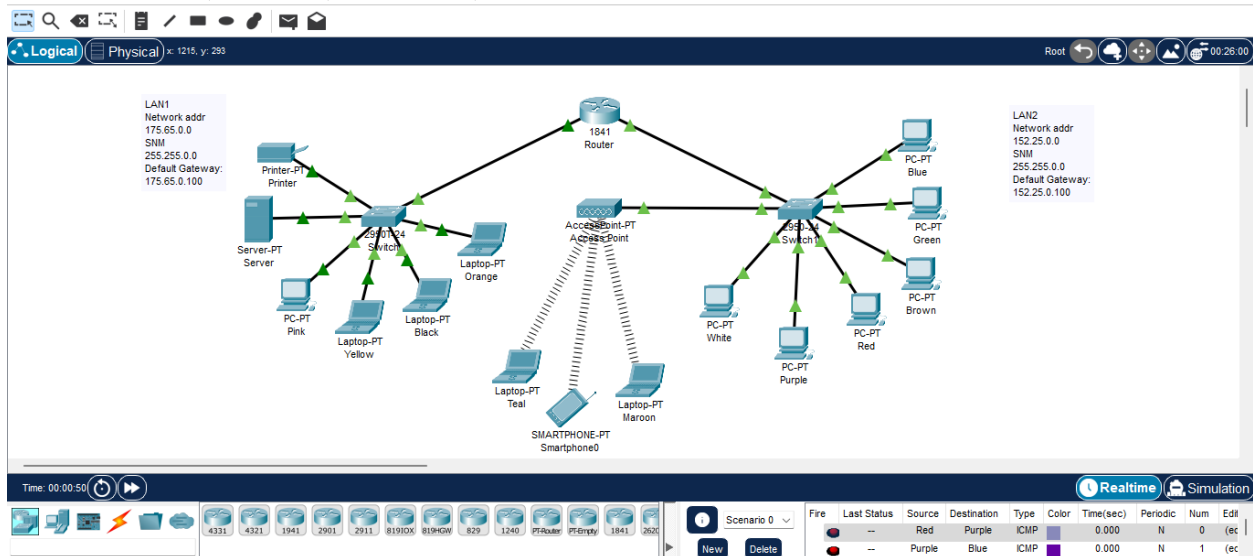


Fig 16. Solution-1

The solution-1 has 6 pcs connected in a LAN, while the second LAN has 4 devices along with a server and printer. Access point is also connected to the 2nd server. The connections are made here using copper cables.

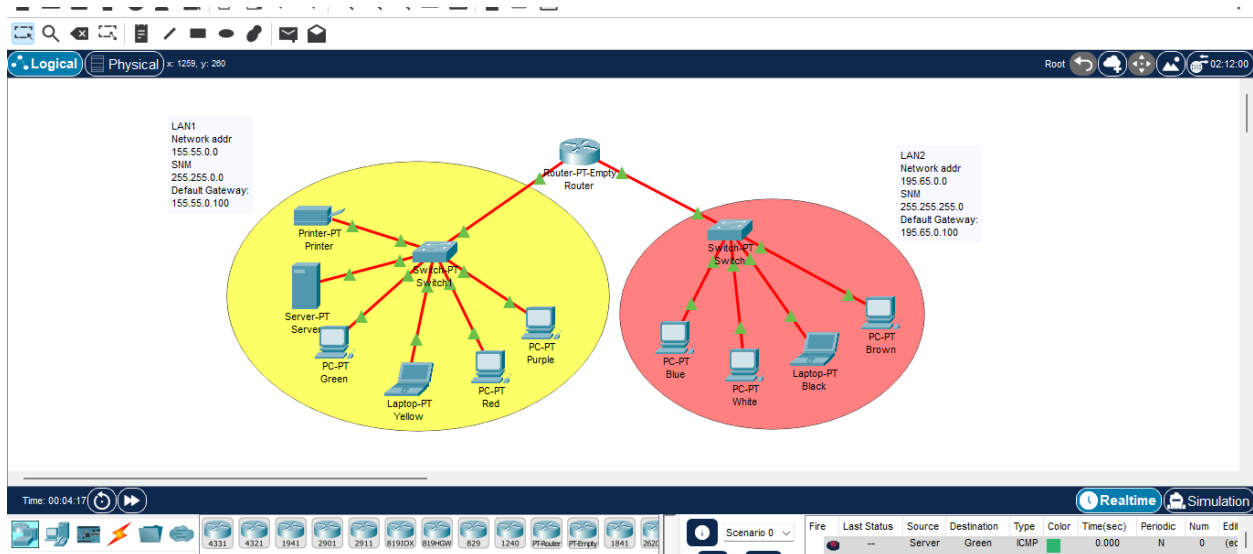


Fig 17. Solution-2

The solution-2 has 4 devices connected in a LAN and another 4 in the LAN1 along with the printer and server. Fiber cables have been used for the connection here.

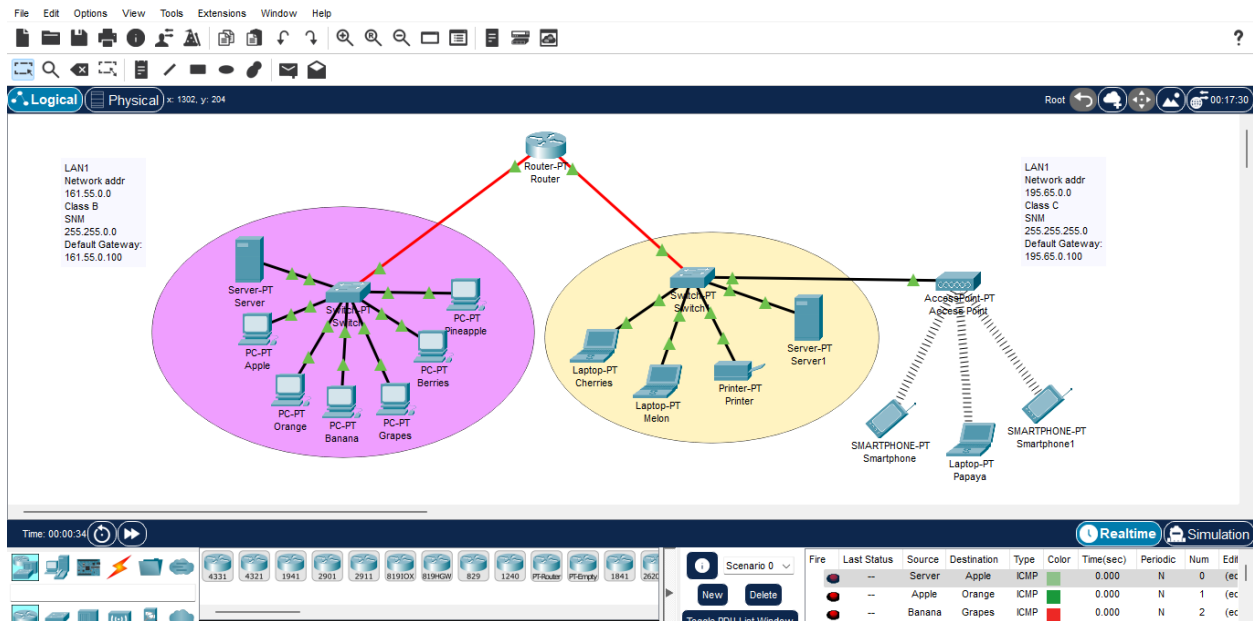


Fig 18. Solution-3

Solution-3 has a mix of copper and fiber cables. It has 2 servers – one handling the email services while other handling the website services. The LAN1 has 6 PCs while the LAN2 has 2 laptops connected. There are printer and an access point connected to LAN2.

TEST PLAN

TESTED BY: Sana Haryani				
SR NO.	TASK NAME	START DATE	END DATE	EXPECTED RESULTS
1.a	Testing Machine Interconnectivity within LAN1	7/3/2022	7/3/2022	Connection Successful.
1.b	Testing Machine Interconnectivity within LAN2	7/3/2022	7/3/2022	Connection Successful.
2.	Testing Machine Interconnectivity between LAN1 and LAN2	7/3/2022	7/3/2022	Connection Successful.
3.	Testing Machine Interconnectivity between printer and pcs.	18/3/2022	18/3/2022	Connection Successful.
4.	Testing interconnectivity between Server and pcs.	18/3/2022	18/3/2022	Connection Successful.
5.	Testing interconnectivity between wired and wireless connection.	18/3/2022	18/3/2022	Connection Successful.
6.	Testing Server website functions – Domain name, etc. (for different LANs)	18/3/2022	18/3/2022	Website on the browser.
7.	Checking mail transfer between different pcs.	18/3/2022	18/3/2022	Mail sent success & Mail receive success.
8.	Checking FTP (upload and transfer file) functions in different machines.	18/3/2022	18/3/2022	File gets added up in the text editor/dir of the machine.

SIMULATION MODE

The interface shows a network diagram on the left with a central switch connected to various devices: Printer-PT, Server-PT, PC-PT (Pink), Laptop-PT (Yellow), Laptop-PT (Black), and Laptop-PT (Orange). The right panel, titled 'Simulation Panel', contains an 'Event List' table, 'Reset Simulation' and 'Constant Delay' buttons, 'Play Controls' (play, stop, fast forward), and 'Event List Filters - Visible Events'.

Vis.	Time(sec)	Last Device	At Device	Type
	0.006	Printer	Switch	ICMP
	0.006	Purple	Switch1	ICMP
	0.006	Switch1	Router	ICMP
	0.006	Switch	Router	ICMP
	0.006	Switch1	Blue	ICMP
	0.006	Pink	Switch	ICMP
	0.006	--	Switch	ICMP
	0.007	Router	Switch1	ICMP
	0.007	Switch	Black	ICMP
	0.007	Switch1	Purple	ICMP

Reset Simulation ☐ Constant Delay Captured to: 0.010 s

Event List Filters - Visible Events
 ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, NDR, NETFLOW, NTP, OSPF, OSPFv6, PaGP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Fig 78. Simulation

The interface is similar to Fig 78, but the 'Event List' table shows events at 0.012 seconds. The network diagram and other UI elements remain the same.

Vis.	Time(sec)	Last Device	At Device	Type
Visible	0.012	--	Switch	ICMP
Visible	0.012	--	Purple	ICMP
Visible	0.012	Switch1	Router	ICMP
Visible	0.012	Switch	Router	ICMP
Visible	0.012	Purple	Switch1	ICMP
Visible	0.012	Switch1	Blue	ICMP
Visible	0.012	Red	Switch1	ICMP
Visible	0.012	Server	Switch	ICMP
Visible	0.012	--	Server	ICMP

Reset Simulation ☐ Constant Delay Captured to: 0.012 s

Event List Filters - Visible Events
 ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, NDR, NETFLOW, NTP, OSPF, OSPFv6, PaGP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Fig 79. Simulation Mode